

ARISTA



CloudVision

CloudVision Workshop 2026

Troubleshooting Section

Section 6 - Troubleshooting Agenda

Using CloudVision to troubleshoot your production network

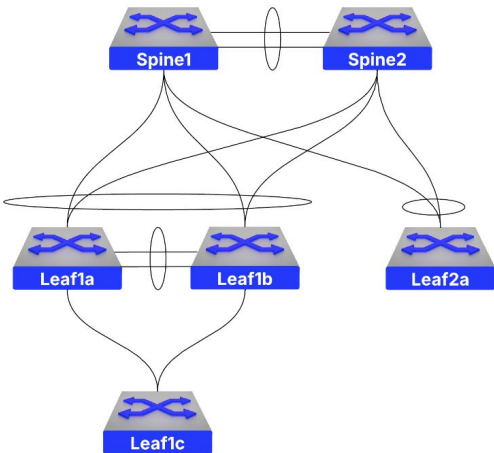
- Time Series
- Device Change tracking and Comparison
- Customized Dashboards

Lab Exercises:

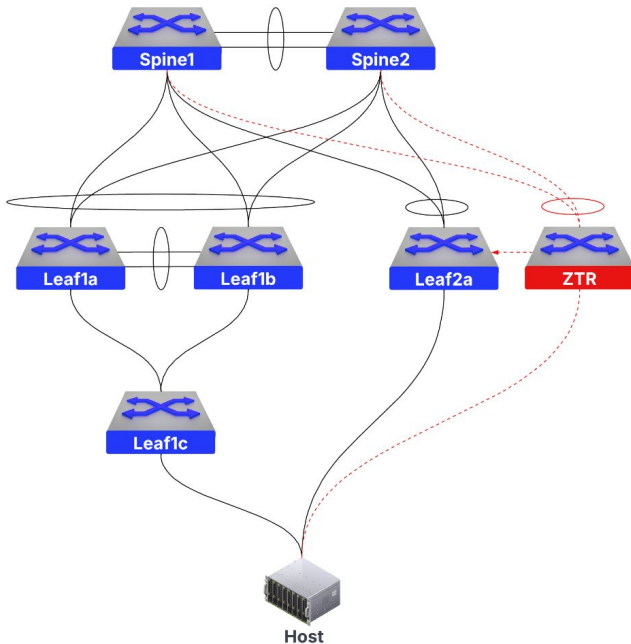
- Lab 6 Section-1 - Configure Connectivity Monitor to probe data plane
- *Instructor initiates Leaf2A failure* 💣
- Lab 6 Section-2 - Replace Leaf2A and migrate config using Zero Touch Replace (ZTR)
- Lab 6 Section-3 - Data Plane packet capture - Use CloudVision to verify connected Host

Section 6 - Lab Topology

Campus A



Campus B



Campus B Leaf2A will undergo replacement

Host is pre-configured with IPs (shown in lab guide)

- Leaf1C connection **10.2.2.1/24**
- Leaf2A connection **10.2.2.2/24**
- Host is running a script to ping from 10.2.2.1 to 10.2.2.2 continuously
- Lab exercise highlights changing GW IP numbering scheme to accommodate

Follow Me - CV Menu Highlights

Viewport Controls

Devices Menu Highlights

- Inventory
- Endpoint Overview
- Compliance - Configuration and Image
- Connectivity Monitor
- Network Utilization 🌟 - *New Summary View!*
- Comparison Demo, by time and device

Events

- Related events example

Network Hierarchy

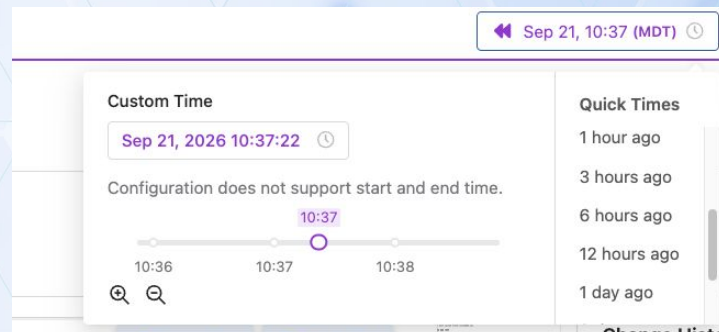
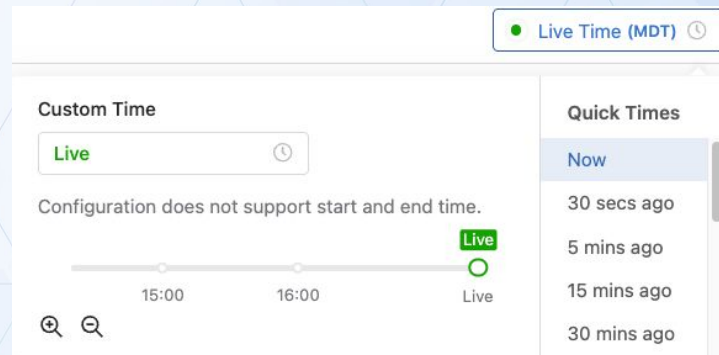
- Drill to alarm
- View topology and events by location

Follow Me - Viewport Navigation

Viewport for real-time and historic views

Selected time follows you around UI to see other telemetry at this point in time

Viewport setting embedded in URL, directly share your exact view



Follow Me - Device Comparison

Time-based Comparison of a Device

- Config deltas
- Snapshots
- Table state

Device Comparison

- Config compare
- Table States



Device at Sep 18, 2026 16:14:23
CampusB-Spine1

Device at Current Time
CampusB-Spine1

Comparing Configuration for CampusB-Spine1 at Sep 18, 2026 16:14:23 against Current Time

Running Config Designed Config

Running Configuration

```
43 interface Port-Channel1
44   description CAMPUSB-LEAF1A_Po1
45   switchport trunk allowed vlan 202
46   switchport mode trunk
47   mlag 1
48 !
49 interface Port-Channel4
50   description CAMPUSB-LEAF2A_Po1
51   switchport trunk allowed vlan 202
52   switchport mode trunk
53   mlag 4
54 !
```

Running Configuration

```
38 interface Port-Channel1
39   description CAMPUSB-LEAF1A_Po1
40
41   switchport mode trunk
42   mlag 1
43 !
44
45 interface Port-Channel3
46   description CAMPUSB-LEAF2A_Po1
47   switchport mode trunk
48   mlag 3
```

Device at Current Time
CampusB-Leaf1A

Device at Current Time
CampusB-Leaf1B

Comparing MAC Address Table for CampusB-Leaf1A against MAC Address Table for CampusB-Leaf1B

VLAN ▼ MAC Address ▼ Internal VLAN ▼ Type ▼ Denylisted ▼ Port ▼ Moves ▼ Last Move ▼

6 Items

Device	VLAN	MAC Address	Internal VLAN	Type	Denylisted	Port	Moves	Last Move
CampusB-Leaf1A	1	02:cd:4b:32:b7:78	No	Dynamic	No	Port-Channel5	1	Sep 21, 2026 16:11:58
CampusB-Leaf1A	1	ae:53:ec:1d:fb:b2	No	Dynamic	No	Port-Channel1	1	Sep 21, 2026 16:11:58
CampusB-Leaf1B	1	02:cd:4b:32:b7:78	No	Dynamic	No	Port-Channel5	1	Sep 21, 2026 16:11:58
CampusB-Leaf1B	1	ae:53:ec:1d:fb:b2	No	Dynamic	No	Port-Channel1	1	Sep 21, 2026 16:11:58

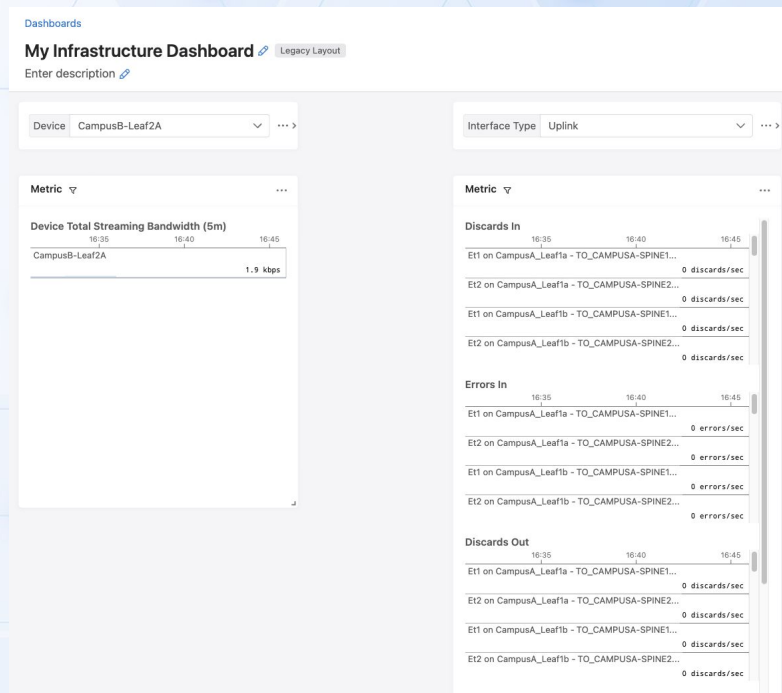
Follow Me - Custom Dashboards

Create dashboard with Device Tag Input Filter Example

- Add a custom device tag to related device
- Use dashboard input to filter by device vs device tag
- *Hint - see additional device-level streaming information*

Create dashboard with Interface Tag Filter Example

- Add custom interface tags to some interesting interfaces
- Add interface tag filter to metrics widget in dashboard
- Can use tags combined with filter by application name, infrastructure roles, PTP types



Instructor Demo - Mirror to CPU / TCPdump

Instructor Pod Steps:

```
CampusB_Leaf1C(config)#monitor session TEST source ethernet 7 rx
```

```
CampusB_Leaf1C(config)#monitor session TEST destination cpu
```

```
CampusB_Leaf1C(config)#bash
```

```
Arista Networks EOS shell
```

```
[arista@CampusB-Leaf1C ~]$ tcpdump -i mirror0
```

```
23:07:35.981758 02:cd:4b:32:b7:78 (oui Unknown) > 00:1c:73:10:02:30 (oui Arista Networks), ethertype IPv4 (0x0800), length 98: 10.2.2.1 > 10.2.2.252: ICMP echo reply, id 32327, seq 2264, length 64  
23:07:36.044857 02:cd:4b:32:b7:78 (oui Unknown) > ae:53:ec:1d:fb:b2 (oui Unknown), ethertype IPv4 (0x0800), length 98: 10.2.2.1 > 10.2.2.2: ICMP echo request, id 1, seq 4660, length 64  
23:07:37.046580 02:cd:4b:32:b7:78 (oui Unknown) > ae:53:ec:1d:fb:b2 (oui Unknown), ethertype IPv4 (0x0800), length 98: 10.2.2.1 > 10.2.2.2: ICMP echo request, id 1, seq 4661, length 64  
23:07:37.982175 02:cd:4b:32:b7:78 (oui Unknown) > 00:1c:73:10:02:30 (oui Arista Networks), ethertype IPv4 (0x0800), length 98: 10.2.2.1 > 10.2.2.252: ICMP echo reply, id 32327, seq 2265, length 64  
23:07:38.047972 02:cd:4b:32:b7:78 (oui Unknown) > ae:53:ec:1d:fb:b2 (oui Unknown), ethertype IPv4 (0x0800), length 98: 10.2.2.1 > 10.2.2.2: ICMP echo request, id 1, seq 4662, length 64  
23:07:38.486894 02:cd:4b:32:b7:78 (oui Unknown) > ae:53:ec:1d:fb:b2 (oui Unknown), ethertype ARP (0x0806), length 42: Request who-has 10.2.2.2 tell 10.2.2.1, length 28  
23:07:38.489968 02:cd:4b:32:b7:78 (oui Unknown) > ae:53:ec:1d:fb:b2 (oui Unknown), ethertype ARP (0x0806), length 42: Reply 10.2.2.1 is-at 02:cd:4b:32:b7:78 (oui Unknown), length 28  
23:07:39.049700 02:cd:4b:32:b7:78 (oui Unknown) > ae:53:ec:1d:fb:b2 (oui Unknown), ethertype IPv4 (0x0800), length 98: 10.2.2.1 > 10.2.2.2: ICMP echo request, id 1, seq 4663, length
```

Instructor provides general best practice guidance for mirroring traffic to CPU vs data plane destinations, engaging TAC, mention DMF including labs.arista.com DMF lab) for scaled implementation.

References including Important limitations by HW platform:

<https://www.arista.com/en/support/toi/eos-4-32-0f/13722-mirror-cpu>

https://arista.my.site.com/AristaCommunity/s/article/introduction-to-managing-eos-devices-troubleshooting#361Configuring_mirroring_to_the_CPU

<https://www.arista.com/en/support/toi/eos-4-20-1f/13935-advanced-mirroring-features#Limitations>

Studios and Tags Troubleshooting

- I'm not getting the config I want example
 - Review config sources menu by device and precedence help article
 - Validate tag matching in studios header
- Recovering workspace submissions
 - Follow-me workspace inputs export/import
- Moving devices between pods, clean up tags
- Workspace management best practices

The screenshot displays the Arista Studios interface for a workspace named 'arista720xpdemo'. At the top, there is a header bar with a green checkmark icon, a green circle icon, and a red circle icon with an exclamation mark. Below the header, the workspace name 'arista720xpdemo (JPE19414328)' is shown. Underneath, there are three tabs: 'Errors' (highlighted with a red circle icon), 'Generated Config' (with a document icon), and 'Logs' (with a document icon). The 'Errors' tab is active, showing a message: 'Studio: Campus Fabric (L2/L3/EVPN)' followed by a red circle icon and the text 'Template error on line 7729 : TagTooManyValuesException:'. Below this, a blue arrow icon points to the text 'Tag error: Access-Pod tag has too many values assigned to device arista720xpdemo, assigned values: manufacturing13, metrology01'.

Studios Settings - Moving Gateway Address

Lab 6 Host testing section

- Preexisting host occupying IP addresses **10.2.2.1/24 & 10.2.2.2/24**
- Studios defaults assign low numbers for router IP
- Solution – Modify **Gateway Address Convention**

Campus Fabric (L2/L3/EVPN) / Advanced Services Settings / Services Allocations

VLAN Gateway Address Convention

Determine which host addresses to use from a VLAN subnet for the gateway address.

☒ First 

☐ Last

Gateway Address Convention

Define the VLAN gateway address convention for a specific set of devices using device tags. This applies to all non-VXLAN VLANs except for a device's in-band management VLAN.

Gateway Address Convention	↓↑	VLAN Gateway Address Convention ⓘ ↓↑
⋮ Campus-Pod: CampusB		Last
+ Add Gateway Address Convention + Add Default Rule		

Troubleshooting Within Network Hierarchy

Default Dashboards

- Self-organizing based on tags assigned by provisioning process
- **Red indicators** on containers with recent alerts
- Topology view is scoped to container selection
- Connectivity Monitor feature summary results displayed
- Status indicators “dwell” a few minutes to indicate to user an event occurred recently—tune using CV Event customization

Compliance

Click-to-resolve from Compliance Menu

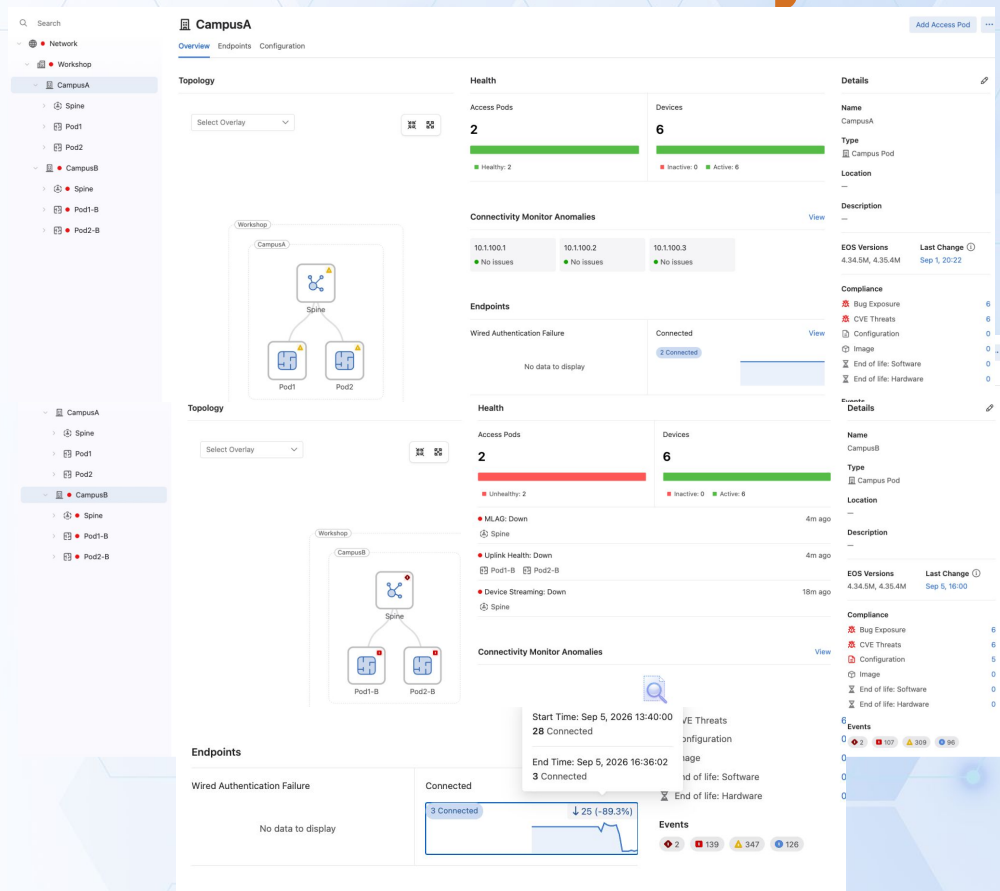
Events Badges

Container Scoped summary of Events

Endpoints

View endpoint changes over time

Represents both network infra and client devices



CloudVision Troubleshooting Fundamentals

“🎵It's your interface to the network data lake🎵”

- 1) **Real-time Streaming Telemetry** - Secure HTTPS/TLS1.3 gRPC Streaming Telemetry of each state change, as it occurs on source devices, not SNMP polling.
- 2) **Time-series** – All the state changes go into time-series.
 - a) **Use MAC, ARP, Route table state change logging**
 - b) **Use Events stream, CV Events are “smart” with high-context mini-dashboards, not just a syslog message**
- 3) **Centralization** – Correlate events from all network layers. Utilize tagging and action templates to create repeatable orchestration rules.
- 4) **Compliance** – Graduated governance journey
 - a) Live health, events, and CVE exposure
 - b) Firmware upgrade orchestration and easy click-to-sync EOS upgrades
 - c) Config management with change-control based synchronization AND reconciliation of manual changes.
 - d) Outside-in automation with CloudVision APIs (Arista AVD, CVPRAC)

Troubleshooting - Automation Callbacks

Inventory - make sure we can pull data from everywhere/everything (no missing objects)

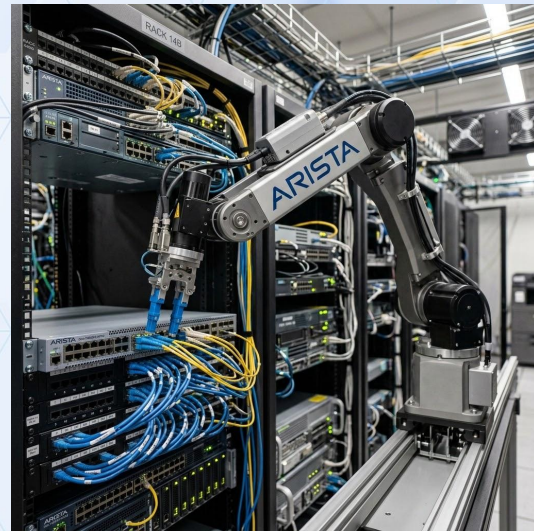
Data - raw data is streamed in and organized within the CV data lake, Telemetry APIs expose this for external access, custom dashboards and query dev tool notebooks built-in

Data Model - make the raw data make sense, inheritance, where to find the scope of impact

Code - In a troubleshooting context, CV telemetry data and config templates must be both machine and human readable.

Proactive (find misconfigs) - use human/machine readable EOS config compliance tools

Reactive (backhoe-as-a-service) - need accurate timestamps events and cascade of failures – correlate with forwarding table state changes



AI-generated image of a Chaos Monkey



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Thank You

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Events Rules - Ignore Subsequent

- Be careful when customizing rule with empty device tag match.
- For rules which alert outside of CloudVision, be sure to explicitly label them and specify reasonable filters so you don't generate SPAM

CloudVision Not Receiving State from Device ⓘ

Event Rules

CloudVision allows you to create custom event rules to get the most useful alerts for generate an event for some devices but not others, or to create events with different

Event rules are checked in the order that they appear here. Rules can be reordered ; used to generate this event.

1

Rule Conditions

Device

ptp

Enable to ignore subsequent rules for selected tags. If no tags are selected, subsequent rules will be disabled for all devices and interfaces

☒ Ignore Subsequent Rules for Selected Tags

CV Onboarding - TerminAttr

- Review HTTPS over TCP connection
- Connection prerequisites
 - MGMT VRF, Gateway, DNS, NTP
 - Token
 - Daemon config string & streaming studio
- Switch onboarding ZTP and post-ZTP behaviors
 - ZTP mode - 24h scheduled reboot
 - Post-ZTP does not do a timer-based rollback (today)
- Checking agent logs - “show agent TerminAttr logs”

TerminAttr Resources

<https://arista.my.site.com/AristaCommunity/s/article/terminattr-most-commonly-used-flags-and-sample-configurations>

<https://www.cv-prod-us-central1-c.arista.io/help/2026.1/articles/provisioning-studios-built-in-streaming-agent#streaming-telemetry-agent-studio>

ZTP Utils:

<https://github.com/aristanetworks/cloudvision-ztpaas-utils>

TA Onboarding Checklist:

1. Create mgmt VRF if using one
2. Management interface (or inband SVI) in correct VRF and with correct IP
3. Default GW in correct VRF (ip route vrf MGMT 0.0.0.0/0 192.168.0.1
4. DNS in correct VRF
5. NTP reachable and servers defined in correct VRF
6. TA string contains correct VRF -cvvrf=<mgmt vrf if not using default>
7. A local admin account is handy during provisioning procedures

Additional Resources

- **Register for Arista TAC Webinar Series:**
<https://events.arista.com/tac-webinar-series>

Previous session recordings -

<https://arista.my.site.com/AristaCommunity/s/article/Arista-TAC-Webinar-Series>

- **Arista Labs On-Demand** <https://labs.arista.com/>

Register and Try it out!

- **Arista Tech Library** - <https://tech-library.arista.com/>

Latest deployment guides and best practices

AVA Prompt Engineering

- AVA built-in limitations:
 - Cannot Configure / edit devices
 - Cannot run commands on switches except specific “show” commands only (high impact commands like `show tech`, `tcpdump`, or `clear bgp *` are not allowed)
- Run command accept - AVA will ask before gathering information from switches
 - You can also establish config context by referencing designed config
 - *Note - AVA does not tolerate underscores in hostnames well today*
- Establishing context through prompting
 - “Use the mac-address table information already within CloudVision to tell me the state of VLAN 202 mac address table on each switch”. Wants to run `show mac address-table vlan 202`
 - Then follow with “Use the CloudVision data lake back-end to tell me the last state change which occurred to VLAN 202 mac table” after context provides useful information. Starting with this prompt alone does not provide the same information.
 - “find devices that have VLAN 202 configured, or search for other types of events that might be related to VLAN 202” – with sufficient context, we get thorough information.

I cannot directly execute `tcpdump` commands as it is not a `show` command and is an interactive command. My capabilities are limited to running read-only `show` commands via eAPI.

If you are trying to diagnose a network issue, I can help by providing information about device configurations, events, or metrics.

① AVA would like to run the following Action:

This action may take longer when run on multiple devices.

⚡ Run Command

Skip

Accept

`show mac address-table vlan 202`



Run on

CampusB_Leaf1A, CampusB_Leaf1B, CampusB_Leaf1C, CampusB_Spine1, +2